COMP0197 Group Coursework
Reproduction Guide

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1 Additional Installed Packages

This project requires three additional packages beyond the standard library, all of which are listed in `requirements.txt`. They can be installed with:

```
pip install torch wfdb matplotlib
```

Package	Used in	Purpose
`torch`	`train.py`, `test.py`, `model.py`, `dataset.py`	Deep learning framework for model definition, training, and inference
`wfdb`	`train.py`, `dataset.py`	Downloading and reading MIT-BIH Arrhythmia Database records
`matplotlib`	`train.py`, `test.py`	Generating loss curves and evaluation plots

Table 1. Additional required packages.

2 Steps to Reproduce All Reported Results

Four configurations are reported. Each requires editing flags in `config.py`, then running `train.py` and `test.py` in sequence. `SEED = 42` in `config.py` defines the base seed; each ensemble member derives its own seed from it (`model_0` uses 42, `model_1` uses 43, `model_2` uses 44).

Step 1: Set Up Environment

Clone the repository and install all packages:

```
git clone https://github.com/Yingjie2004422/COMP0197_GW.git
cd COMP0197_GW
pip install -r requirements.txt
```

To install manually instead:

```
pip install torch wfdb matplotlib
```

The MIT-BIH Arrhythmia Database (48 records, 360 Hz) is downloaded automatically on the first run of `train.py`; no manual download is needed.

Step 2: Full Model (Default)

Verify the following flags in `config.py`:

```
USE_ATTENTION = True
USE_RR_FEATURES = True
USE_RISK_HEAD = True
DETERMINISTIC = False
MODEL_DIR = 'checkpoints_default'
RESULTS_DIR = 'results_default'
```

Then run:

```
python train.py
python test.py
```

Expected: Signal MSE = 0.957, AUC-ROC = 0.903, Winkler Score = 5.293.

Step 3: Ablation — No Attention

Set `USE_ATTENTION = False` in `config.py`:

```
USE_ATTENTION = False
MODEL_DIR = 'checkpoints_no_attention'
RESULTS_DIR = 'results_no_attention'
```

```
python train.py
python test.py
```

Expected: Signal MSE = 0.991, AUC-ROC = 0.935, ECE = 0.451.

Step 4: Ablation — No RR Features

```
USE_ATTENTION = True
USE_RR_FEATURES = False
MODEL_DIR = 'checkpoints_no_RR'
RESULTS_DIR = 'results_no_RR'
```

```
python train.py
python test.py
```

Expected: Signal MSE = 1.414, AUC-ROC = 0.919, Winkler Score = 5.914.

Step 5: Ablation — Deterministic Baseline

```
USE_ATTENTION = True
USE_RR_FEATURES = True
DETERMINISTIC = True
MODEL_DIR = 'checkpoints_deterministic'
RESULTS_DIR = 'results_deterministic'
```

```
python train.py
python test.py
```

Expected: Signal MSE = 1.000, AUC-ROC = 0.947, Winkler Score = 5.013.

3 Results Verification

After running all four configurations, check the key metrics in each `test.log` against Table 2.

Configuration	Log file	MSE	AUC	F1	Winkler
Full model	`results_default/test.log`	0.957	0.903	0.683	5.293
No attention	`results_no_attention/test_no_attention.log`	0.991	0.935	0.665	5.202
No RR features	`results_no_RR/test_no_RR.log`	1.414	0.919	0.734	5.914
Deterministic	`results_deterministic/test_deterministic.log`	1.000	0.947	0.697	5.013

Table 2. Expected metrics for each configuration.

Notes

Seed. `SEED = 42` in `config.py` defines the base seed. Each ensemble member uses a different seed derived from it: `model_0` uses seed 42, `model_1` uses 43, and `model_2` uses 44 (`seed = SEED + model_idx`).

Hardware. Training was run on a CUDA GPU. CPU runs give the same results but are roughly 5–10× slower per epoch.

Data download. `train.py` calls `wfdb.dl_database` automatically if the `mitdb/` folder is absent. Internet access is required on the first run (≈110 MB).